Physics 5C Midterm 1 Review Session

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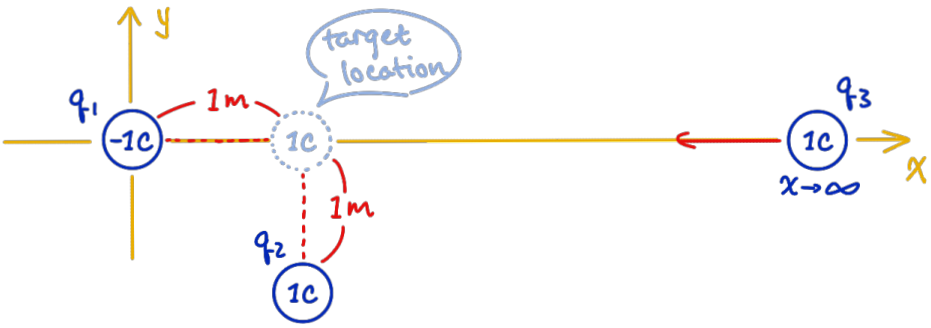
Consider the following configuration:
A charge $q_1 = -1\,\mathrm{C}$ is fixed at $\vec{x}_1 = (0, 0)\,\mathrm{m}$.
A charge $q_2 = 1\,\mathrm{C}$ is fixed at $\vec{x}_2 = (1, -1)\,\mathrm{m}$.
Jinyan is trying to bring in a test charge $q_3 = 1\,\mathrm{C}$ from infinitely far away at $\vec{x}_i = (\infty, 0)\,\mathrm{m}$ to the target location at $\vec{x}_f = (1, 0)\,\mathrm{m}$. We will discuss this process from two perspectives, first using electric potential and energy, then using electric field and forces.



First we consider the picture of electric potential.
Part (a) Compute the electric potential at $\vec{x}_i$ and $\vec{x}_f$ due to q_1 and q_2 combined.
Part (b) Compute the electric potential energy stored in q_3 at $\vec{x}_i$ and $\vec{x}_f$.
Part (c) How much energy is needed to bring q_3 from $\vec{x}_i$ to $\vec{x}_f$?
Part (d) If q_3 is fixed at $\vec{x}_f = (1, 0)\,\mathrm{m}$, compute the potential energy stored in q_2.
Part (e) What is the total energy needed to assemble the three-charge system? [1]

Now let us consider the picture of electric field.
Part (f) Draw the electric field lines due to the source charges q_1 and q_2 combined.
Part (g) Compute the net electric field due to q_1 and q_2 combined, at $\vec{x}_i$ and at $\vec{x}_f$.
Part (h) Compute the net force that q_3 experiences when it is at $\vec{x}_i$ and at $\vec{x}_f$.
Part (i) Jinyan flips the sign of q_3 when it is at $\vec{x}_f$. What gets changed in this picture?

Now we explore the relation between electric field and electric potential.
Consider only the source charges q_1 and q_2, that is, neglect the test charge q_3.
Part (j) Find the location (other than very far away) where we have the "net" electric potential being zero. Pick one of those locations and compute the magnitude of the electric field at that location. Qualitatively describe the relation between electric field and electric potential.

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[1] That is, this should be a three-step calculation: Step 1 - Starting with a vacuum, nothing in space, find the energy to put q_1 at the origin $\vec{x}_1$; Step 2 - With only q_1 fixed at the origin, find the energy needed to bring in q_2 from infinity to $\vec{x}_2 = (1, -1)\,\mathrm{m}$; Step 3 - With q_1 and q_2 fixed at their locations, find the energy needed to bring in q_3 from infinity to $\vec{x}_f$. Sum these energies to find the total energy stored in such a system.